HARD ALGEBRA

Systems of two linear equations in two variables

01

10 questions · ~15 min

SAT QUESTION BANK

2026-05-29

Q1 GRID-IN ALGEBRA

$$y = 4x + 1$$
$$4y = 15x - 8$$

The solution to the given system of equations is x, y. What is the value of $x - y$?

STUDENT-PRODUCED RESPONSE

Write the answer above. Fill the matching bubble for each digit, plus the fraction bar or decimal point if needed.

Q2 ALGEBRA

$$4x - 9y = 9y + 5$$

$$hy = 2 + 4x$$

In the given system of equations, h is a constant. If the system has no solution, what is the value of h?

A. -9

B. 0

C. 9

D. 18

Q3 ALGEBRA

$$ax + by = 72$$

$$6x + 2by = 56$$

In the given system of equations, a and b are constants. The graphs of these equations in the xy-plane intersect at the point $4, y$. What is the value of a?

A. 3

B. 4

C. 6

D. 14

Q4 ALGEBRA

Which system of linear equations has no solution?

A. $-2x + 3y = -9$
$2x - 3y = 9$

B. $2x - 3y = 9$
$3x + 4y = 10$

C. $2x - 3y = 9$
$-6x + 9y = -27$

D. $-2x + 3y = 9$
$4x - 6y = 18$

Q5 ALGEBRA

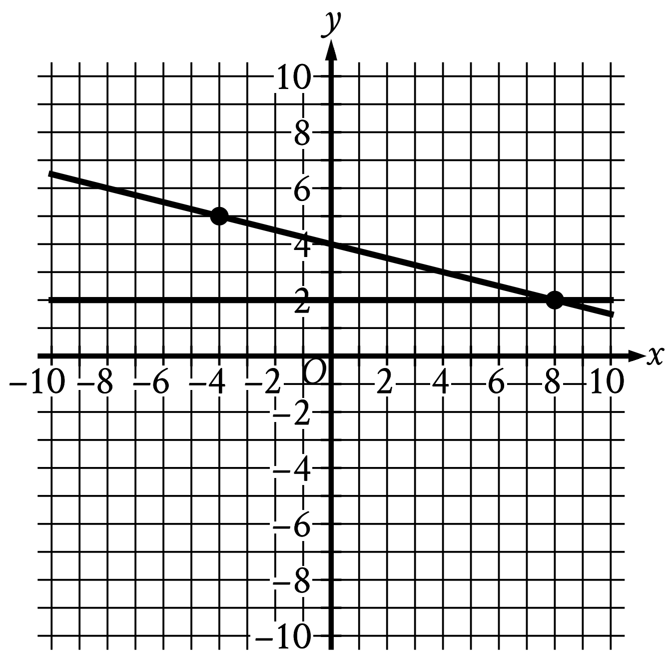
Store A sells raspberries for $\$5.50$ per pint and blackberries for $\$3.00$ per pint. Store B sells raspberries for $\$6.50$ per pint and blackberries for $\$8.00$ per pint. A certain purchase of raspberries and blackberries would cost $\$37.00$ at Store A or $\$66.00$ at Store B. How many pints of blackberries are in this purchase?

A. 4

B. 5

C. 8

D. 12



- For the first line in the system:
  - The line slants gradually down from left to right.
  - The line passes through the following points:
    - (negative 4 comma 5)
    - (0 comma 4)
    - (8 comma 2)
- For the second line in the system:
  - The line is horizontal.
  - The line passes through the following points:
    - (0 comma 2)
    - (8 comma 2)

If a new graph of three linear equations is created using the system of equations shown and the equation $x + 4y = -16$, how many solutions x, y will the resulting system of three equations have?

A. Zero

B. Exactly one

C. Exactly two

D. Infinitely many

Q7

GRID-IN ALGEBRA

$$6 + 7r = pw$$

$$7r - 5w = 5w + 11$$

In the given system of equations, p is a constant. If the system has no solution, what is the value of p?

STUDENT-PRODUCED RESPONSE

.	/ .	/ .	.
0	0	0	0
1	1	1	1
2	2	2	2
3	3	3	3
4	4	4	4
5	5	5	5
6	6	6	6
7	7	7	7
8	8	8	8
9	9	9	9

Write the answer above. Fill the matching bubble for each digit, plus the fraction bar or decimal point if needed.

Q8

GRID-IN ALGEBRA

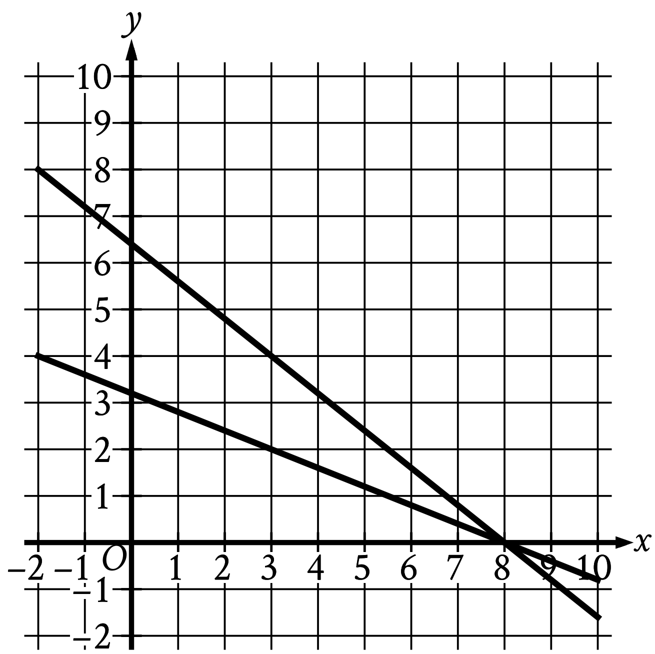
$$x - 2 - 4y + 7 = 117$$

$$x - 2 + 4y + 7 = 442$$

The solution to the given system of equations is x, y. What is the value of $6x - 2$?

STUDENT-PRODUCED RESPONSE

Write the answer above. Fill the matching bubble for each digit, plus the fraction bar or decimal point if needed.



- For the first line in the system:
  - The line slants gradually down from left to right.
  - The line passes through the following points:
    - (3 comma 2)
    - (8 comma 0)
- For the second line in the system:
  - The line slants gradually down from left to right.
  - The line passes through the following points:
    - (3 comma 4)
    - (8 comma 0)

What system of linear equations is represented by the lines shown?

A. $8x + 4y = 32$
$-10x - 4y = -64$

B. $8x - 4y = 32$

$$-10x + 4y = -64$$

C. $4x - 10y = 32$
$-8x + 10y = -64$

D. $4x + 10y = 32$
$-8x - 10y = -64$

Q10 GRID-IN ALGEBRA

$$48x - 72y = 30y + 24$$

$$ry = \frac{1}{6} - 16x$$

In the given system of equations, r is a constant. If the system has no solution, what is the value of r?

STUDENT-PRODUCED RESPONSE

Write the answer above. Fill the matching bubble for each digit, plus the fraction bar or decimal point if needed.